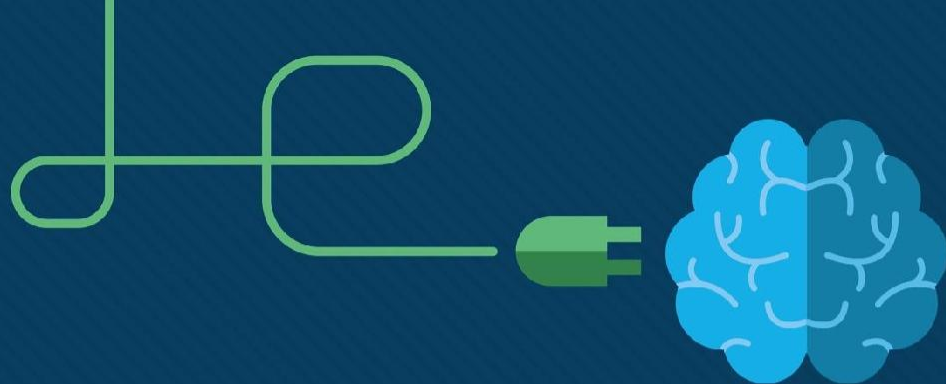




Chapter 6: IPv4 Addressing

Introduction to Networks v7.02
(ITN)



Objectives

Title: IPv4 Addressing

Objective: Calculate an IPv4 subnetting scheme to efficiently segment your network.

Topic Title	Topic Objective
IPv4 Address Structure	Describe the structure of an IPv4 address including the network portion, the host portion, and the subnet mask.
IPv4 Unicast, Broadcast, and Multicast	Compare the characteristics and uses of the unicast, broadcast and multicast IPv4 addresses.
Types of IPv4 Addresses	Explain public, private, and reserved IPv4 addresses.
Network Segmentation	Explain how subnetting segments a network to enable better communication.
Subnet an IPv4 Network	Calculate IPv4 subnets for a /24 prefix.

Objectives (Cont.)

Title: IPv4 Addressing

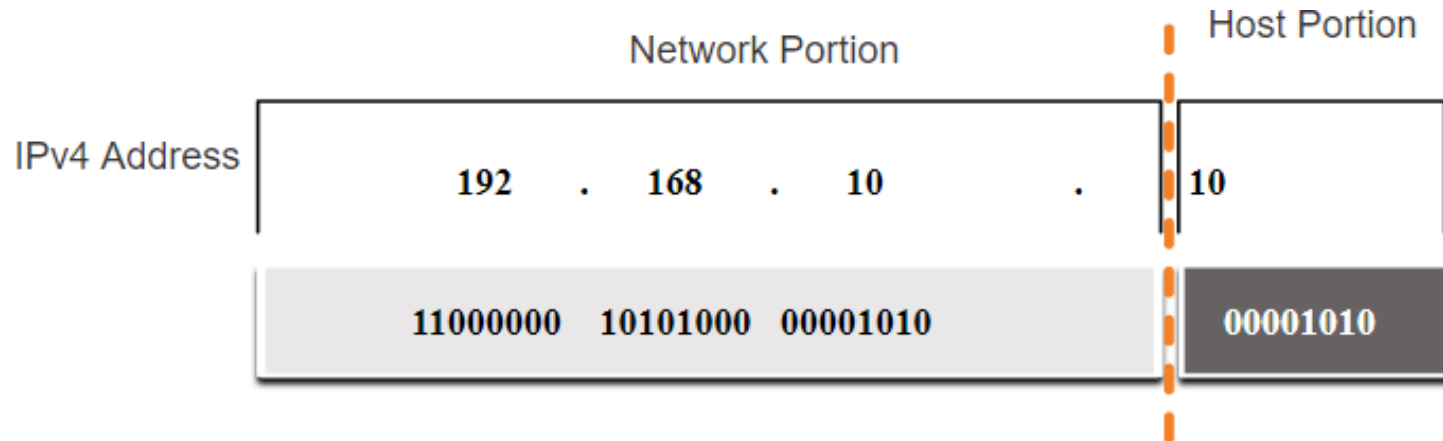
Objective: Calculate an IPv4 subnetting scheme to efficiently segment your network.

Topic Title	Topic Objective
Subnetting a /16 and a /8 Prefix	Calculate IPv4 subnets for a /16 and a /8 prefix.
Subnetting to Meet Requirements	Given a set of requirements for subnetting, implement an IPv4 addressing scheme.
Variable Length Subnet Masking (VLSM)	Explain how to create a flexible addressing scheme using variable length subnet masking (VLSM).
Structured Design	Implement a VLSM addressing scheme.

6.1 IPv4 Address Structure

Network and Host Portions

- An IPv4 address is a 32-bit hierarchical address that is made up of a **network portion** and a **host portion**.
- When determining the network portion versus the host portion, you must look at the 32-bit stream.
- All devices on the **same network** must have the **identical network portion**.
- A **subnet mask** is used to **determine the network and host portions**.



The Subnet Mask (Cont.)

- The IPv4 address is compared to the subnet mask bit by bit, from left to right.
- A 1 in the subnet mask indicates that the corresponding bit in the IPv4 address is a network bit.

	Network Portion			Host Portion	
IPv4 Address	192	.	168	.	10
	11000000	10101000	00001010	00001010	
Subnet Mask	255	.	255	.	0
	11111111	11111111	11111111	00000000	

IPv4 Address Structure

The Subnet Mask

- To identify the network and host portions of an IPv4 address, the subnet mask is compared to the IPv4 address bit for bit, from left to right.
- The actual process used to identify the network and host portions is called ANDing.

	Network Portion					Host Portion
IPv4 Address	192	.	168	.	10	10
	11000000	10101000	00001010			00001010
Subnet Mask	255	.	255	.		0
	11111111	11111111	11111111			00000000

IPv4 Address Structure

The Prefix Length

- A prefix length is a less cumbersome method **used to identify a subnet mask address.**

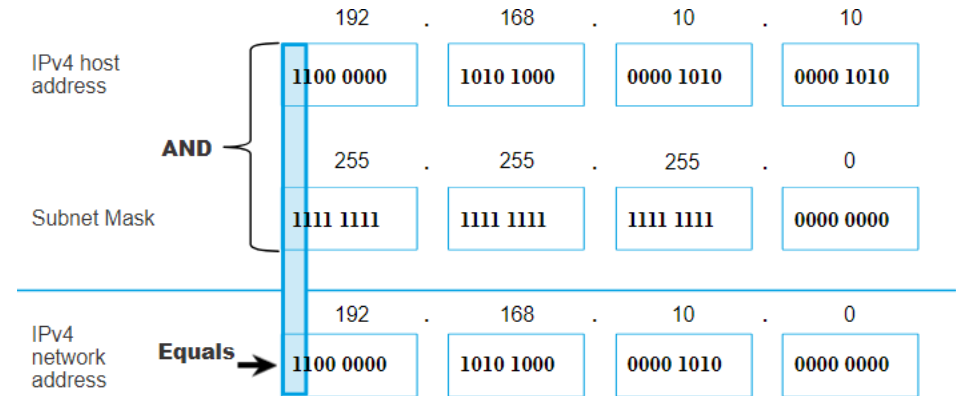
- The prefix length is the number of bits set to 1 in the subnet mask.

- It is written in “slash notation” therefore, count the number of bits in the subnet mask and prepend it with a slash.

Subnet Mask	32-bit Address	Prefix Length
255.0.0.0	11111111.00000000.00000000.00000000	/8
255.255.0.0	11111111.11111111.00000000.00000000	/16
255.255.255.0	11111111.11111111.11111111.00000000	/24
255.255.255.128	11111111.11111111.11111111.10000000	/25
255.255.255.192	11111111.11111111.11111111.11000000	/26
255.255.255.224	11111111.11111111.11111111.11100000	/27
255.255.255.240	11111111.11111111.11111111.11110000	/28
255.255.255.248	11111111.11111111.11111111.11111000	/29
255.255.255.252	11111111.11111111.11111111.11111100	/30

Determining the Network: Logical AND

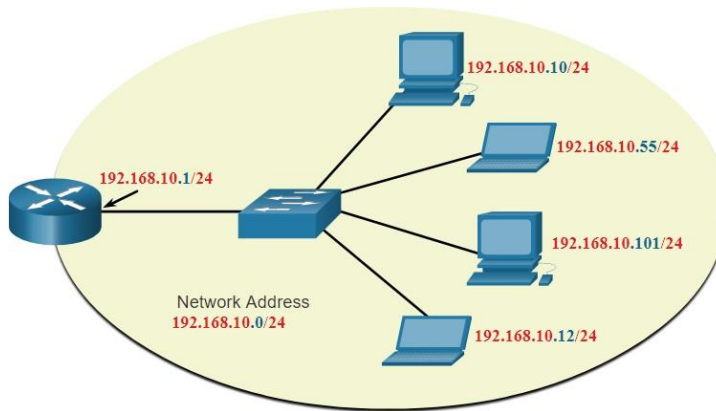
- A **logical AND** Boolean operation is used in determining the network address.
- Logical AND is the comparison of two bits where only a 1 AND 1 produces a 1 and any other combination results in a 0.
- $1 \text{ AND } 1 = 1$, $0 \text{ AND } 1 = 0$, $1 \text{ AND } 0 = 0$, $0 \text{ AND } 0 = 0$
- $1 = \text{True}$ and $0 = \text{False}$
- To identify the network address, the host IPv4 address is logically ANDed, bit by bit, with the subnet mask to identify the network address.



IPv4 Address Structure

Network, Host, and Broadcast Addresses

- Within each network are three types of IP addresses:
 - Network address
 - Host addresses
 - Broadcast address

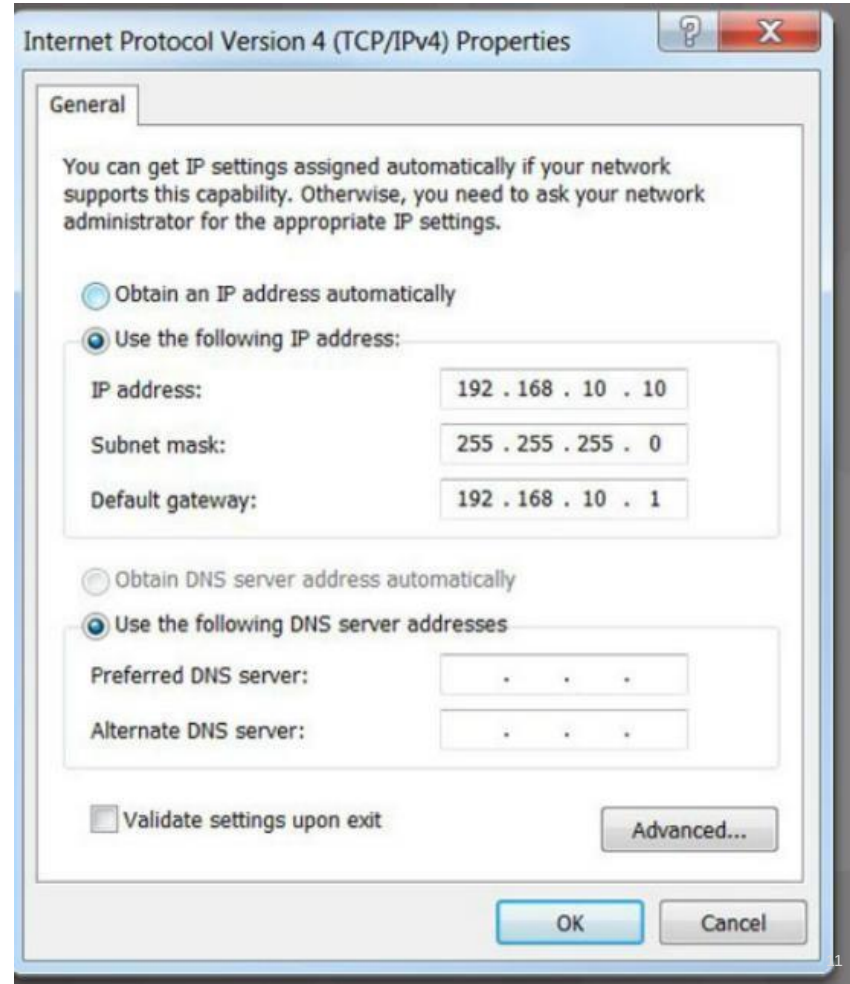


	Network Portion			Host Portion	Host Bits
Subnet mask 255.255.255.0 or /24	255	255	255	0	
	11111111	11111111	11111111	00000000	
Network address 192.168.10.0 or /24	192	168	10	0	All 0s
	11000000	10100000	00001010	00000000	
First address 192.168.10.1 or /24	192	168	10	1	All 0s and a 1
	11000000	10100000	00001010	00000001	
Last address 192.168.10.254 or /24	192	168	10	254	All 1s and a 0
	11000000	10100000	00001010	11111110	
Broadcast address 192.168.10.255 or /24	192	168	10	255	All 1s
	11000000	10100000	00001010	11111111	

IPv4 Address Structure

The Subnet Mask

- Three IPv4 addresses must be configured on a host:
 - **Unique IPv4** address of the host.
 - **Subnet mask** - identifies the network/host portion of the IPv4 address.
 - **Default gateway** -IP address of the local router interface.



6.2 IPv4 Unicast, Broadcast, and Multicast

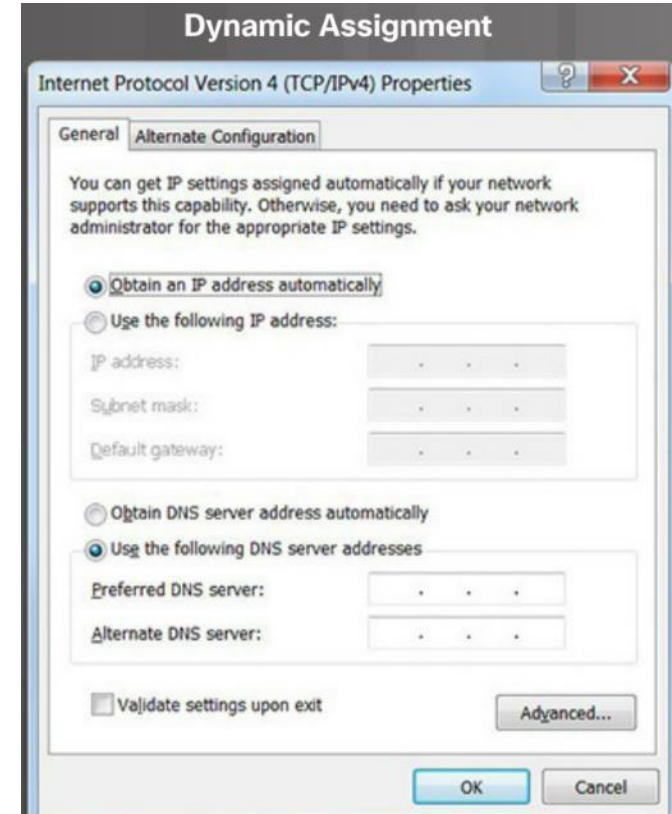
Static IPv4 Address Assignment to a Host

- Some devices like printers, servers and network devices require a fixed IP address.
- Hosts in a small network can also be configured with static addresses.



Dynamic IPv4 Address Assignment to a Host

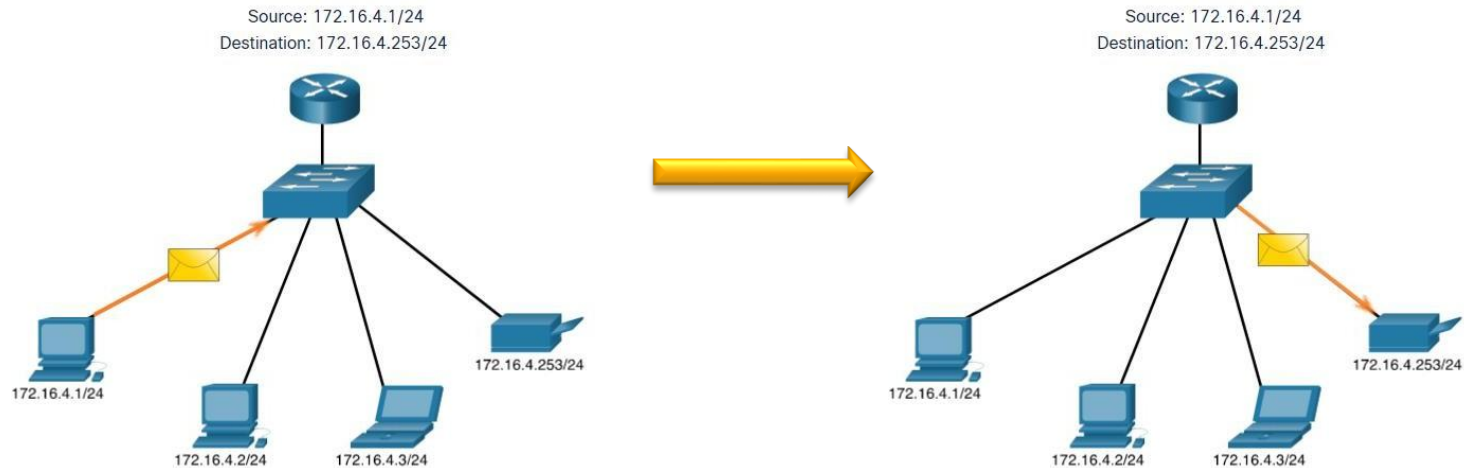
- Most networks use **Dynamic Host Configuration Protocol (DHCP)** to assign IPv4 addresses dynamically.
- The DHCP server provides an **IPv4 address, subnet mask, default gateway, and other configuration information.**
- DHCP **leases the addresses to hosts** for a certain length of time.
- If the host is powered down or taken off the network, the address is returned to the pool for reuse.



IPv4 Unicast, Broadcast, and Multicast

Unicast

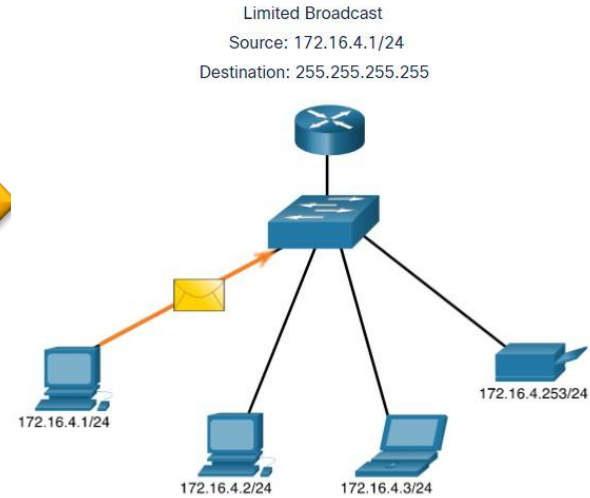
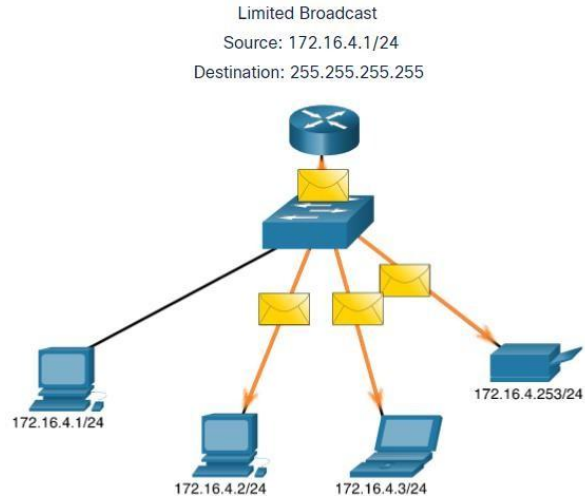
- Unicast transmission is sending a packet to one destination IP address.
- For example, the PC at 172.16.4.1 sends a unicast packet to the printer at 172.16.4.253.



IPv4 Unicast, Broadcast, and Multicast

Broadcast

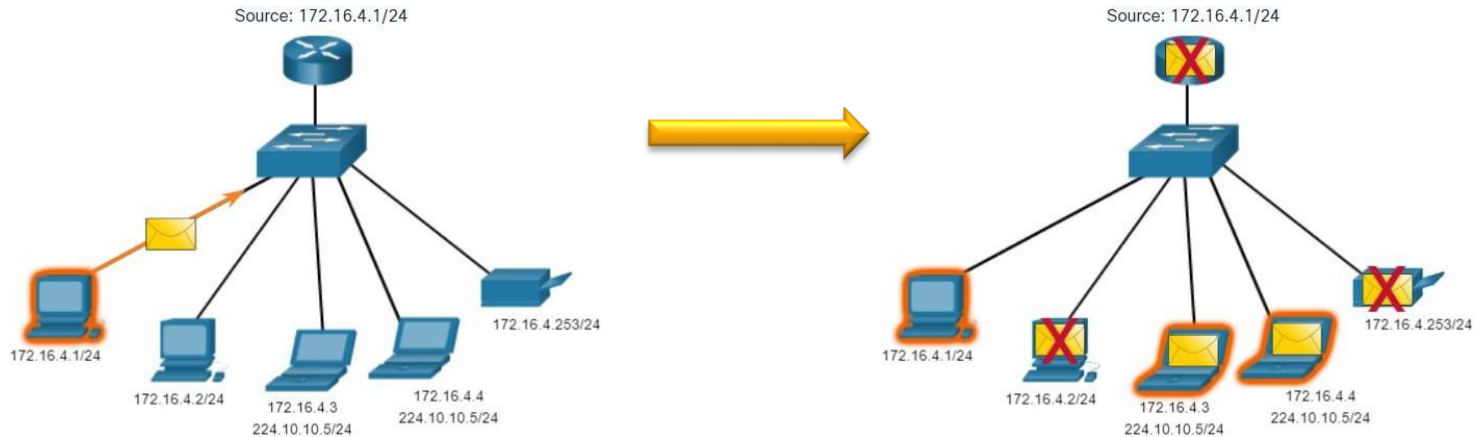
- Broadcast transmission is sending a packet to all other destination IP addresses.
- For example, the PC at 172.16.4.1 sends a broadcast packet to all IPv4 hosts.



IPv4 Unicast, Broadcast, and Multicast

Multicast

- Multicast transmission is sending a packet to a multicast address group.
- For example, the PC at 172.16.4.1 sends a multicast packet to the multicast group address 224.10.10.5.



6.3 Types of IPv4 Addresses

Public and Private IPv4 Addresses

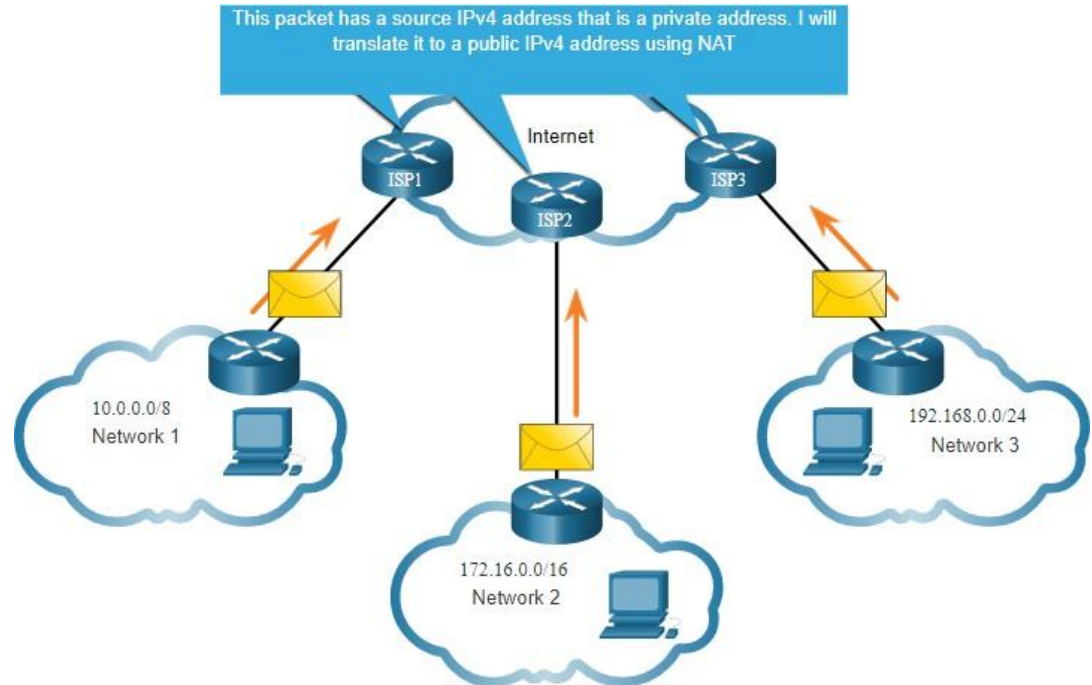
- As defined in in RFC 1918, **public IPv4 addresses are globally routed between internet service provider (ISP) routers.**
- Private addresses are common blocks of addresses used by most organizations to assign IPv4 addresses to internal hosts.
- **Private IPv4 addresses are not unique and can be used internally within any network.**
- However, private addresses are not globally routable.

Network Address and Prefix	RFC 1918 Private Address Range
10.0.0.0/8	10.0.0.0 - 10.255.255.255
172.16.0.0/12	172.16.0.0 - 172.31.255.255
192.168.0.0/16	192.168.0.0 - 192.168.255.255

Types of IPv4 Addresses

Routing to the Internet

- Network Address Translation (NAT) translates private IPv4 addresses to public IPv4 addresses.
- NAT is typically enabled on the edge router connecting to the internet.
- It translates the internal private address to a public global IP address.



Special Use IPv4 Addresses

Loopback addresses

- 127.0.0.0 /8 (127.0.0.1 to 127.255.255.254)
- Commonly identified as only 127.0.0.1
- Used on a host to test if TCP/IP is operational.

```
C:\Users\NetAcad> ping 127.0.0.1
Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
```

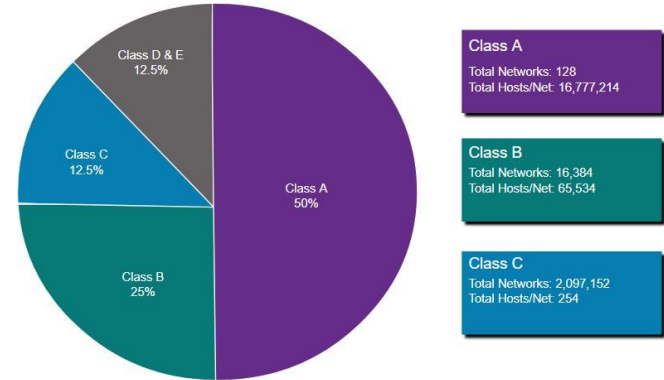
Link-Local addresses

- 169.254.0.0 /16 (169.254.0.1 to 169.254.255.254)
- Commonly known as the Automatic Private IP Addressing (APIPA) addresses or self-assigned addresses.
- Used by Windows DHCP clients to self-configure when no DHCP servers are available.

Legacy Classful Addressing

RFC 790 (1981) allocated IPv4 addresses in classes

- Class A (0.0.0.0/8 to 127.0.0.0/8)
 - Class B (128.0.0.0 /16 – 191.255.0.0 /16)
 - Class C (192.0.0.0 /24 – 223.255.255.0 /24)
 - Class D (224.0.0.0 to 239.0.0.0)
 - Class E (240.0.0.0 – 255.0.0.0)
-
- Classful addressing wasted many IPv4 addresses.



Classful address allocation was replaced with classless addressing which ignores the rules of classes (A, B, C).

Legacy Classful Addressing

IP Address Classes

Address Class	1st octet range (decimal)	1st octet bits (green bits do not change)	Network(N) and Host(H) parts of address	Default subnet mask (decimal and binary)	Number of possible networks and hosts per network
A	1-127**	00000000-01111111	N.H.H.H	255.0.0.0	128 nets (2^7) 16,777,214 hosts per net (2^{24-2})
B	128-191	10000000-10111111	N.N.H.H	255.255.0.0	16,384 nets (2^{14}) 65,534 hosts per net (2^{16-2})
C	192-223	11000000-11011111	N.N.N.H	255.255.255.0	2,097,150 nets (2^{21}) 254 hosts per net (2^{8-2})
D	224-239	11100000-11101111	NA (multicast)		
E	240-255	11110000-11111111	NA (experimental)		

Legacy Classful Addressing

- In 1981, Internet IPv4 addresses were assigned using classful addressing (RFC 790)
- Network addresses were based on 3 classes:
 - **Class A** (0.0.0.0/8 to 127.0.0.0/8) – Designed to support extremely large networks with more than 16 million host addresses.

Class A Specifics	
Address Block	0.0.0.0 – 127.0.0.0
Default Subnet Mask	/8 (255.0.0.0)
Maximum Number of Networks	128
Number of Host per Network	16,777,214
High order bit	0xxxxxxx.____.____.____

* 0.0.0.0 and 127.0.0.0 are reserved and cannot be assigned

Legacy Classful Addressing

- **Class B** (128.0.0.0 /16 – 191.255.0.0 /16) – Designed to support the needs of moderate to large size networks up to approximately 65,000 host addresses.

Class B Specifics	
Address Block	128.0.0.0 - 191.255.0.0
Default Subnet Mask	/16 (255.255.0.0)
Maximum Number of Networks	16,384
Number of Host per Network	65,534
High order bit	10xxxxxx.____.____.____

Legacy Classful Addressing

- **Class C** (192.0.0.0 /24 – 223.255.255.0 /24) – Designed to support small networks with a maximum of 254 hosts.

Class C Specifics	
Address Block	192.0.0.0 - 223.255.255.0
Default Subnet Mask	/24 (255.255.255.0)
Maximum Number of Networks	2,097,152
Number of Host per Network	254
High order bit	110xxxxx.____.____.____

Types of IPv4 Addresses

Assignment of IP Addresses

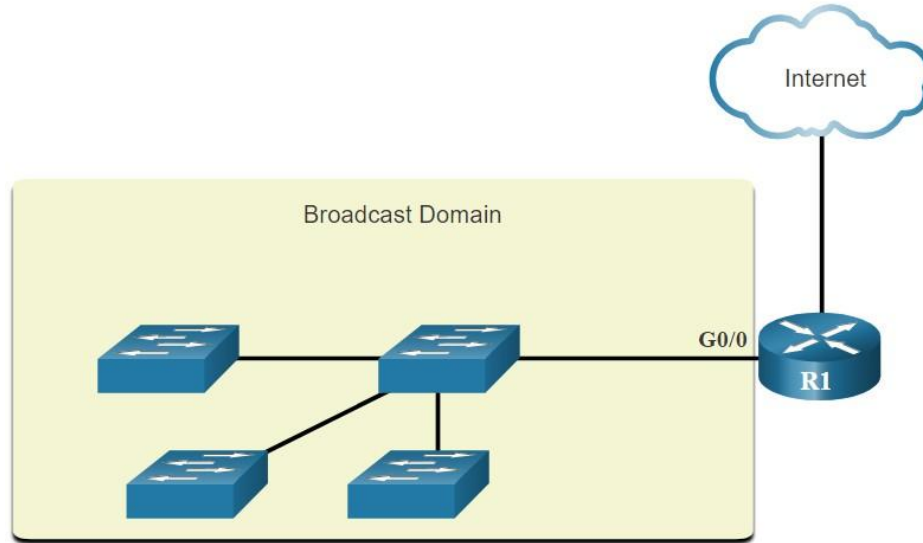
- The Internet Assigned Numbers Authority (IANA) manages and allocates blocks of IPv4 and IPv6 addresses to five Regional Internet Registries (RIRs).
- RIRs are responsible for allocating IP addresses to ISPs who provide IPv4 address blocks to smaller ISPs and organizations.



Network Segmentation

Broadcast Domains and Segmentation

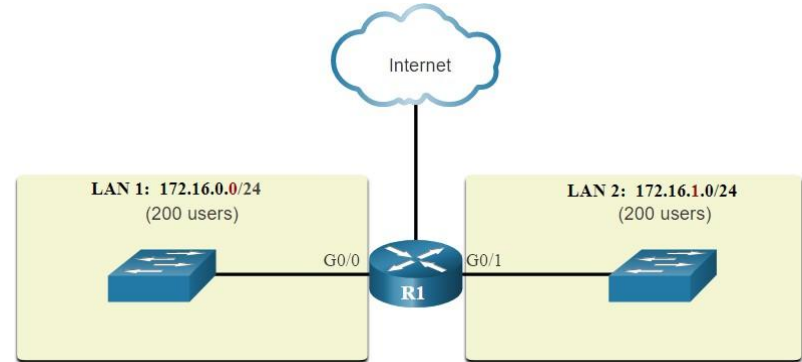
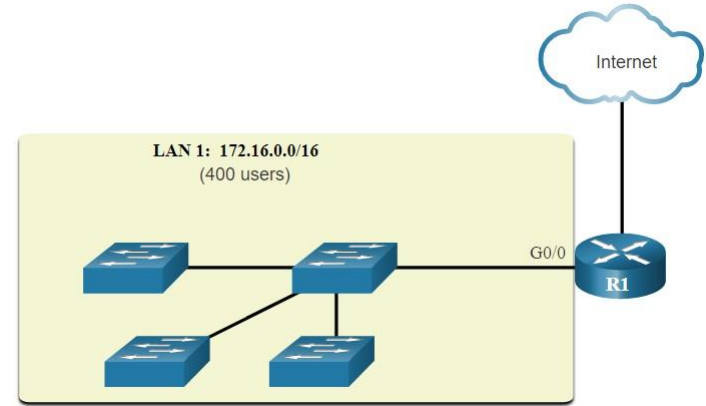
- Many protocols use broadcasts or multicasts (e.g., ARP use broadcasts to locate other devices, hosts send DHCP discover broadcasts to locate a DHCP server.)
- Switches propagate broadcasts out all interfaces except the interface on which it was received.



- The only device that stops broadcasts is a router.
- Routers do not propagate broadcasts.
- Each router interface connects to a broadcast domain and broadcasts are only propagated within that specific broadcast domain.

Problems with Large Broadcast Domains

- A problem with a large broadcast domain is that these hosts can generate excessive broadcasts and negatively affect the network.
- The solution is to reduce the size of the network to create smaller broadcast domains in a process called subnetting.
- Dividing the network address 172.16.0.0 /16 into two subnets of 200 users each: 172.16.0.0 /24 and 172.16.1.0 /24.
- Broadcasts are only propagated within the smaller broadcast domains.

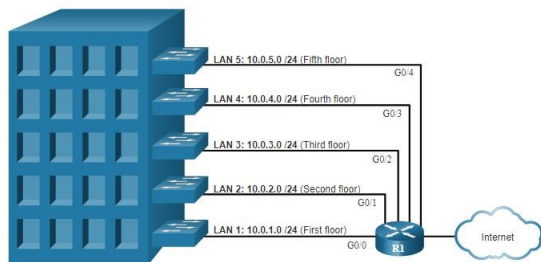


Network Segmentation

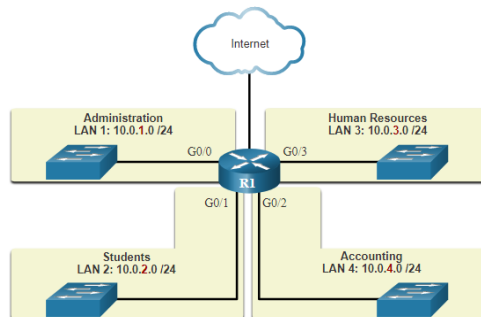
Reasons for Segmenting Networks

- Subnetting reduces overall network traffic and improves network performance.
 - It can be used to implement security policies between subnets.
 - Subnetting reduces the number of devices affected by abnormal broadcast traffic.
- Subnets are used for a variety of reasons including by:

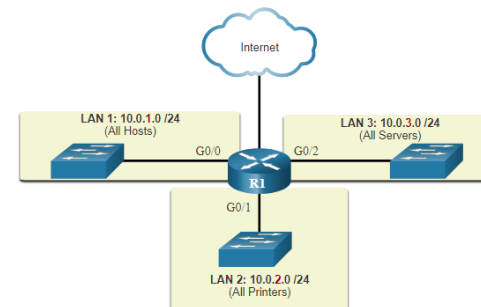
Location



Group or Function



Device Type



6.4 Subnet an IPv4 Network

Subnet on an Octet Boundary

- Networks are most easily subnetted at the octet boundary of /8, /16, and /24.
- Notice that using longer prefix lengths decreases the number of hosts per subnet.

Subnets are created by borrowing host bits for network bits
More host bits borrowed, the more subnets that can be defined

Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of hosts
/8	255.0.0.0	<code>nnnnnnnnn.hhhhhhhh.hhhhhhhh.hhhhhhhh</code> <code>11111111.00000000.00000000.00000000</code>	16,777,214
/16	255.255.0.0	<code>nnnnnnnnn.nnnnnnnnn.hhhhhhhh.hhhhhhhh</code> <code>11111111.11111111.00000000.00000000</code>	65,534
/24	255.255.255.0	<code>nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.hhhhhhhh</code> <code>11111111.11111111.11111111.00000000</code>	254

Subnetting an IPv4 Network • 10.0.0.0/8 is subnetted using /16 mask.

Subnetting on the Octet Boundary

Subnet Address (256 Possible Subnets)	Host Range (65,534 possible hosts per subnet)	Broadcast
<u>10.0.0.0/16</u>	<u>10.0.0.1</u> – <u>10.0.255.254</u>	<u>10.0.255.255</u>
<u>10.1.0.0/16</u>	<u>10.1.0.1</u> – <u>10.1.255.254</u>	<u>10.1.255.255</u>
<u>10.2.0.0/16</u>	<u>10.2.0.1</u> – <u>10.2.255.254</u>	<u>10.2.255.255</u>
<u>10.3.0.0/16</u>	<u>10.3.0.1</u> – <u>10.3.255.254</u>	<u>10.3.255.255</u>
<u>10.4.0.0/16</u>	<u>10.4.0.1</u> – <u>10.4.255.254</u>	<u>10.4.255.255</u>
<u>10.5.0.0/16</u>	<u>10.5.0.1</u> – <u>10.5.255.254</u>	<u>10.5.255.255</u>
<u>10.6.0.0/16</u>	<u>10.6.0.1</u> – <u>10.6.255.254</u>	<u>10.6.255.255</u>
<u>10.7.0.0/16</u>	<u>10.7.0.1</u> – <u>10.7.255.254</u>	<u>10.7.255.255</u>
...	...	...
<u>10.255.0.0/16</u>	<u>10.255.0.1</u> – <u>10.255.255.254</u>	<u>10.255.255.255</u>

- Subnetting Network 10.x.0.0/16
- Define up to 256 subnets with each subnet capable of connecting 65,534 hosts.
- First two octets identify the network portion while the last two octets are for host IP addresses.

- 10.0.0.0/8 is subnetted using a /24 mask.

Subnetting on the Octet Boundary (Cont.)

Subnet Address (65,536 Possible Subnets)	Host Range (254 possible hosts per subnet)	Broadcast
10.0.0.0/24	10.0.0.1 - 10.0.0.254	10.0.0.255
10.0.1.0/24	10.0.1.1 - 10.0.1.254	10.0.1.255
10.0.2.0/24	10.0.2.1 - 10.0.2.254	10.0.1.255
...	...	...
10.0.255.0/24	10.0.255.1 - 10.0.255.254	10.0.255.255
10.1.0.0/24	10.1.0.1 - 10.1.0.254	10.1.0.255
10.1.1.0/24	10.1.1.1 - 10.1.1.254	1.1.1.0.255
10.1.2.0/24	10.1.2.1 - 10.1.2.254	10.1.2.0.255
...	...	...
10.100.0.0/24	10.100.0.1 - 10.100.0.254	10.100.0.255
...	...	...
10.255.255.0/24	10.255.255.1 - 10.255.255.254	10.255.255.255

- Subnetting Network 10.x.x.0/24
- Define 65,536 subnets each capable of connecting 254 hosts.
- /24 boundary is very popular in subnetting because of number of hosts.

Classless Subnetting

Subnetting a /24 Network

Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of subnets	# of hosts
/25	255.255.255.128	n n n n n n n n . n n n n n n n n . n n n n n n n n . n h h h h h h h 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 0 0 0 0 0 0 0	2	126
/26	255.255.255.192	n n n n n n n n . n n n n n n n n . n n n n n n n n . n n h h h h h h 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 0 0 0 0 0 0	4	62
/27	255.255.255.224	n n n n n n n n . n n n n n n n n . n n n n n n n n . n n n h h h h h 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 0 0 0 0 0	8	30
/28	255.255.255.240	n n n n n n n n . n n n n n n n n . n n n n n n n n . n n n n h h h h 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 0 0 0 0	16	14
/29	255.255.255.248	n n n n n n n n . n n n n n n n n . n n n n n n n n . n n n n n h h h 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 0 0 0	32	6
/30	255.255.255.252	n n n n n n n n . n n n n n n n n . n n n n n n n n . n n n n n n h h 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 0 0	64	2

Subnets can borrow bits from *any* host bit position to create other masks.

Video Demonstration – The Subnet Mask

Subnetting in Binary

▪ ANDING

- Convert IP address and Subnet Mask to Binary (line up vertically like an addition problem)
- Logically AND (1 and 1 = 1, all other combinations = 0)
- Result is network address for original IP address

▪ Classful Subnetting

- Class A /8 255.0.0.0
- Class B /16 255.255.0.0
- Class C /24 255.255.255.0



Video Demonstration – The Subnet Mask (Cont.)

Subnetting 192.168.1.0/24

192	168	1	0
255	255	255	128
11000000	10101000	00000001	00000000
11111111	11111111	11111111	10000000
N	N	N	SN H

Subnet bits = $2^1 = 2$
Host bits = $2^7 = 128 - 2 = 126$
Subnetworks = 2



Subnetting 192.168.1.0/24

192	168	1	68
255	255	255	128
11000000	10101000	00000001	01000100
11111111	11111111	11111111	10000000
11000000	10101000	00000001	00000000
192	168	1	0

192.168.1.0 /25 -----> 192.168.1.127 /25
192.168.1.128 /25 -----> 192.168.1.255 /25

Video Demonstration – Subnetting with the Magic Number

- Magic number technique used to calculate subnets
- Magic number is simply the place value of the last one in the subnet mask
- /25 11111111.11111111.11111111.10000000 magic number = 128
- /26 11111111.11111111.11111111.11000000 magic number = 64
- /27 11111111.11111111.11111111.11100000 magic number = 32



Video – Subnet with the Magic Number

-



Demonstration

Subnetting with the Magic Number

Subnetting an IPv4 Network

Classless Subnetting Example

192.168.1.0/25 Network

Borrow 1 bit from the host portion of the address.

Original	192.	168.	1.	0	000 0000	1 Network
Mask	255.	255.	255.	0	000 0000	

The borrowed bit value is 0 for the Net 0 address.

Net 0	192.	168.	1.	0	000 0000	2 Subnets
Net 1	192.	168.	1.	1	000 0000	

The new subnets have the SAME subnet mask.

Mask	255.	255.	255.	1	000 0000
------	------	------	------	---	----------

Dotted Decimal Addresses

Borrow 1 bit from the host portion of the address.

Original	192.	168.	1.	0	000 0000	1 Network
Mask	255.	255.	255.	0	000 0000	

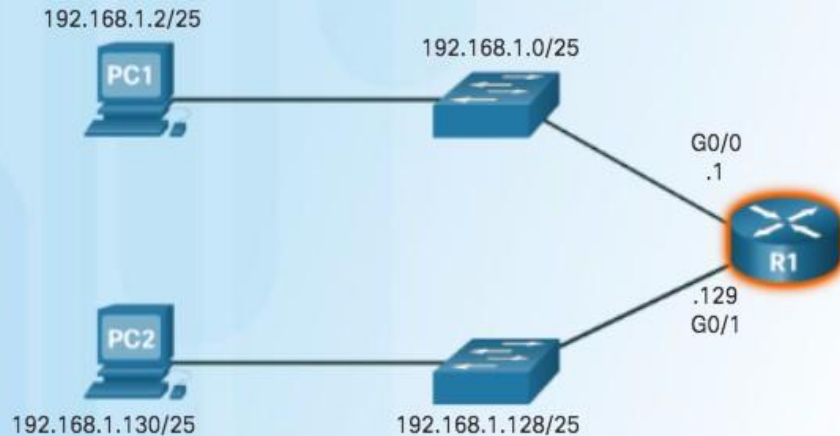
	192.	168.	1.	0/25		
Net 0	192.	168.	1.	0	000 0000	2 Subnets
Net 1	192.	168.	1.	128/25		

	255.	255.	255.	128	
Mask	255.	255.	255.	1	000 0000

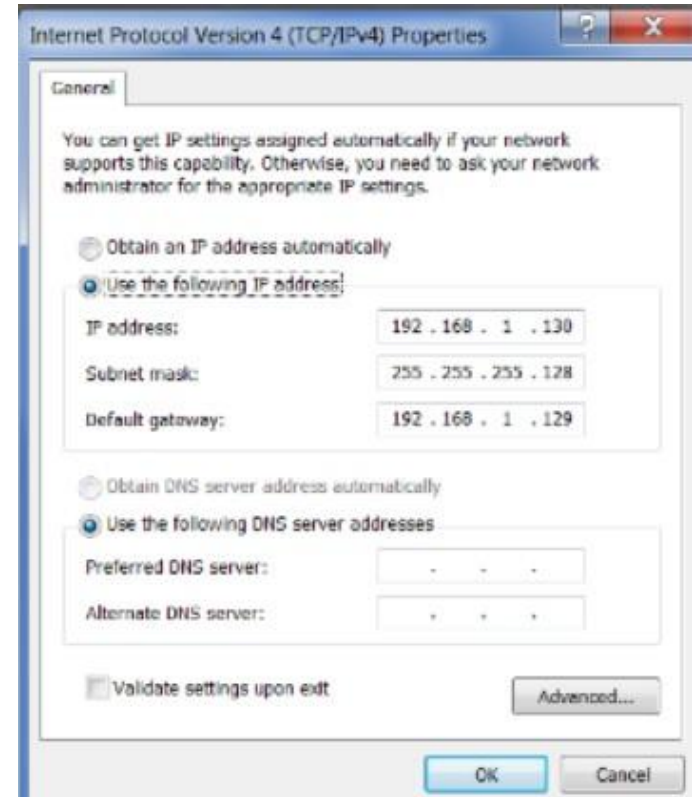
Subnetting an IPv4 Network

Creating 2 Subnets

▪ /25 Subnetting Topology



```
R1(config)# interface gigabitethernet 0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.128
R1(config-if)# exit
R1(config)# interface gigabitethernet 0/1
R1(config-if)# ip address 192.168.1.129 255.255.255.128
```



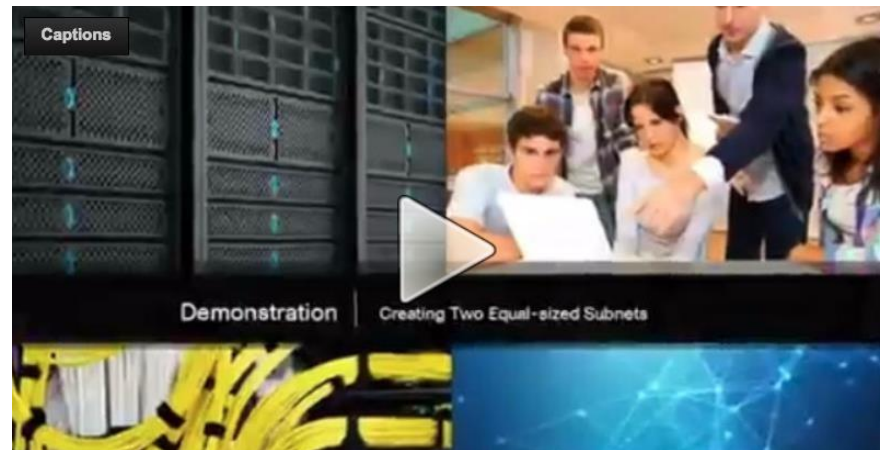
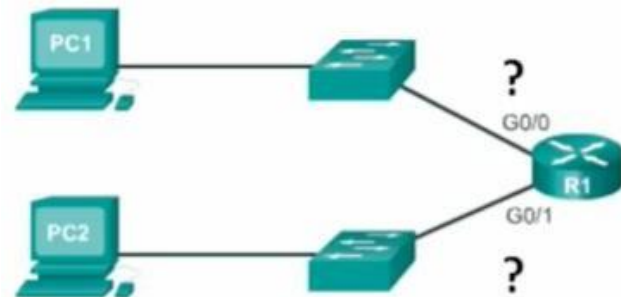
Video Demonstration – Creating Two Equal-sized Subnets (/25)

Create 2 Equal-sized Subnets from 192.168.1.0 /24

- **Subnet Mask** - 11111111.11111111.11111111.10000000

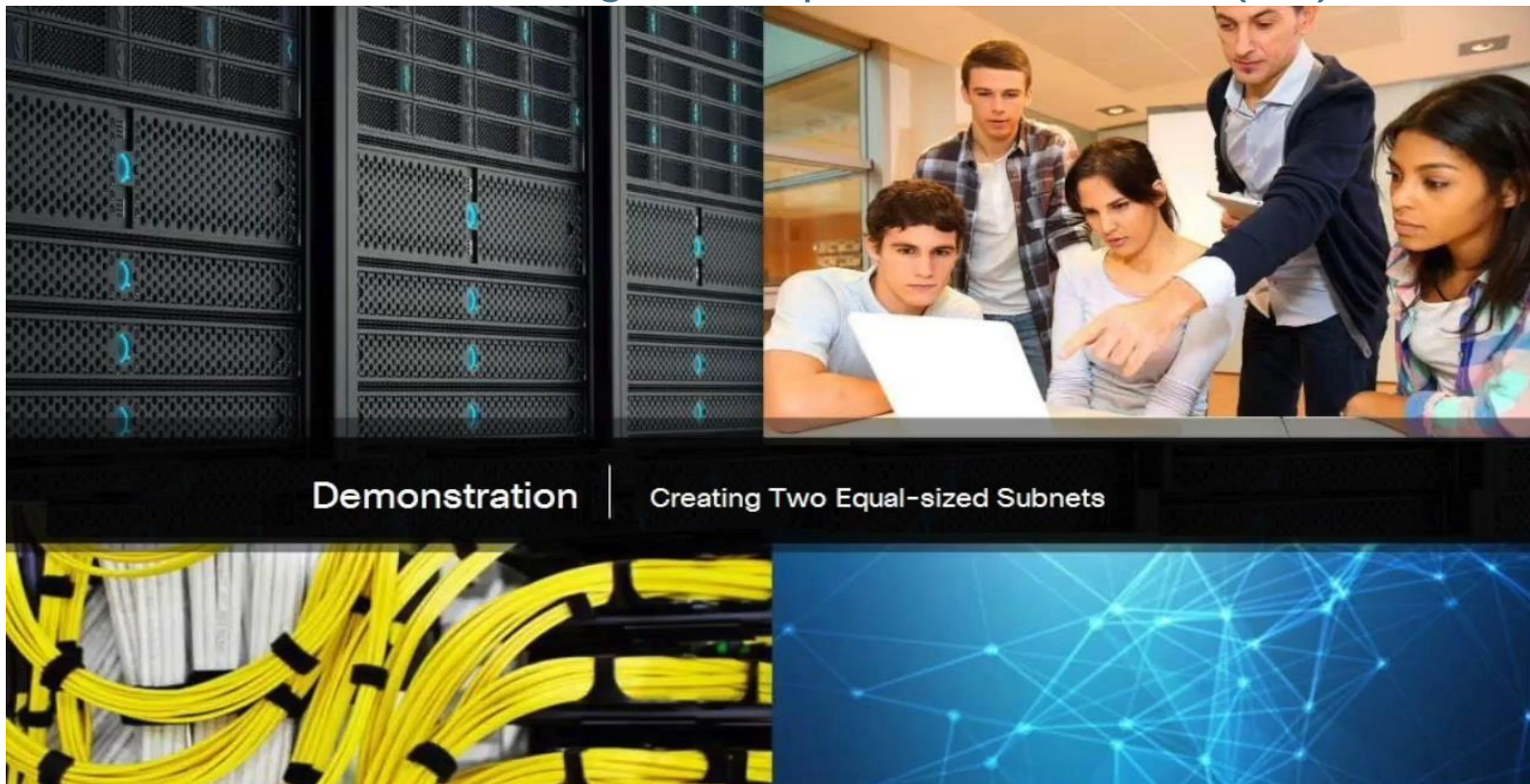
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
1	0	0	0	0	0	0	0

- Magic Number = 128
- 192.168.1.0 /25 (start at 0)
- 192.168.1.128 /25 (Add 128)



Creating 2 Subnets – Video

Video Demonstration – Creating Two Equal-sized Subnets (/25)



Subnetting an IPv4 Network

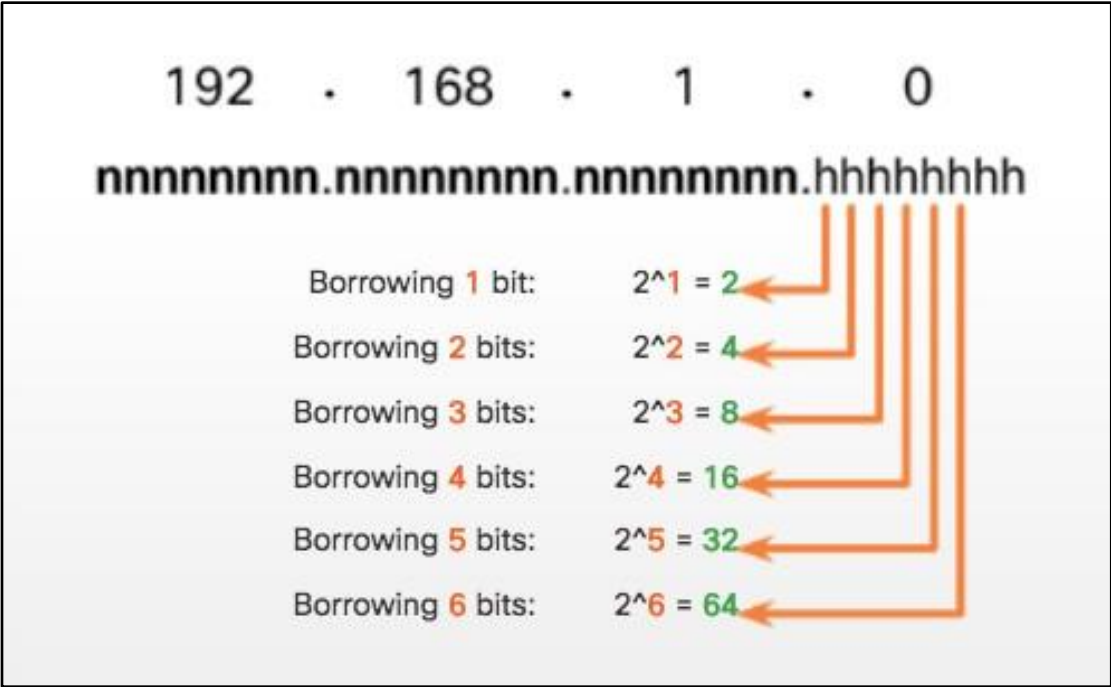
Subnetting Formulas

Calculate Number of Subnets Formula

$$2^n$$

n = bits borrowed

Subnetting a /24 Network



Subnetting Formulas (Cont.)

Calculate Number of Hosts Formula

$$2^n - 2$$

n = the number of bits remaining in the host field

Calculating the Number of Hosts

192.168.1.0

0000000

7 bits remain in host field

$2^7 = 128$ IP addresses per subnet
 $2^7 - 2 = 126$ host IP addresses per subnet

Subnet a Slash 16 and a Slash 8 Prefix

Subnet a Slash 16 and a Slash 8 Prefix

Create Subnets with a Slash 16 prefix

- The table highlights all the possible scenarios for subnetting a /16 prefix.

Prefix Length	Subnet Mask	Network Address (n = network, h = host)	# of subnets	# of hosts
/17	255.255.128.0	nnnnnnnn.nnnnnnnn.nhshhshhsh.hshhshhshh 11111111.11111111.10000000.00000000	2	32766
/18	255.255.192.0	nnnnnnnn.nnnnnnnn.nnhshhshhsh.hshhshhshh 11111111.11111111.11000000.00000000	4	16382
/19	255.255.224.0	nnnnnnnn.nnnnnnnn.nnnshhshhsh.hshhshhshh 11111111.11111111.11100000.00000000	8	8190
/20	255.255.240.0	nnnnnnnn.nnnnnnnn.nnnnshhshh.hshhshhshh 11111111.11111111.11110000.00000000	16	4094
/21	255.255.248.0	nnnnnnnn.nnnnnnnn.nnnnnshhsh.hshhshhshh 11111111.11111111.11111000.00000000	32	2046
/22	255.255.252.0	nnnnnnnn.nnnnnnnn.nnnnnnshh.hshhshhshh 11111111.11111111.11111100.00000000	64	1022
/23	255.255.254.0	nnnnnnnn.nnnnnnnn.nnnnnnnsh.hshhshhshh 11111111.11111111.11111110.00000000	128	510
/24	255.255.255.0	nnnnnnnn.nnnnnnnn.nnnnnnnnn.hshhshhshh 11111111.11111111.11111111.00000000	256	254
/25	255.255.255.128	nnnnnnnn.nnnnnnnn.nnnnnnnnn.nhshhshhshh 11111111.11111111.11111111.10000000	512	126
/26	255.255.255.192	nnnnnnnn.nnnnnnnn.nnnnnnnnn.nnhshhshhshh 11111111.11111111.11111111.11000000	1024	62
/27	255.255.255.224	nnnnnnnn.nnnnnnnn.nnnnnnnnn.nnnshhshhshh 11111111.11111111.11111111.11100000	2048	30
/28	255.255.255.240	nnnnnnnn.nnnnnnnn.nnnnnnnnn.nnnnshhshhshh 11111111.11111111.11111111.11110000	4096	14
/29	255.255.255.248	nnnnnnnn.nnnnnnnn.nnnnnnnnn.nnnnnshhshhshh 11111111.11111111.11111111.11111000	8192	6
/30	255.255.255.252	nnnnnnnn.nnnnnnnn.nnnnnnnnn.nnnnnnshhshhshh 11111111.11111111.11111111.11111100	16384	2



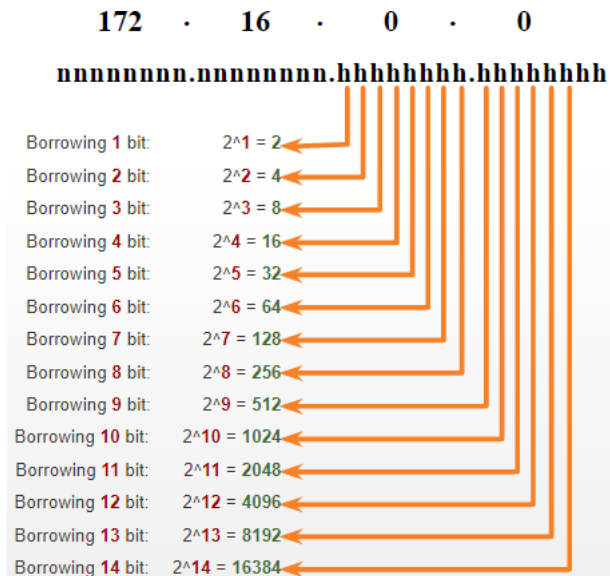
Subnet a Slash 16 and a Slash 8 Prefix

Create 100 Subnets with a Slash 16 prefix

Consider a large enterprise that requires at least 100 subnets and has chosen the private address 172.16.0.0/16 as its internal network address.

- The figure displays the number of subnets that can be created when borrowing bits from the third octet and the fourth octet.
- Notice there are now up to 14 host bits that can be borrowed (i.e., last two bits cannot be borrowed).

To satisfy the requirement of 100 subnets for the enterprise, 7 bits (i.e., $2^7 = 128$ subnets) would need to be borrowed (for a total of 128 subnets).



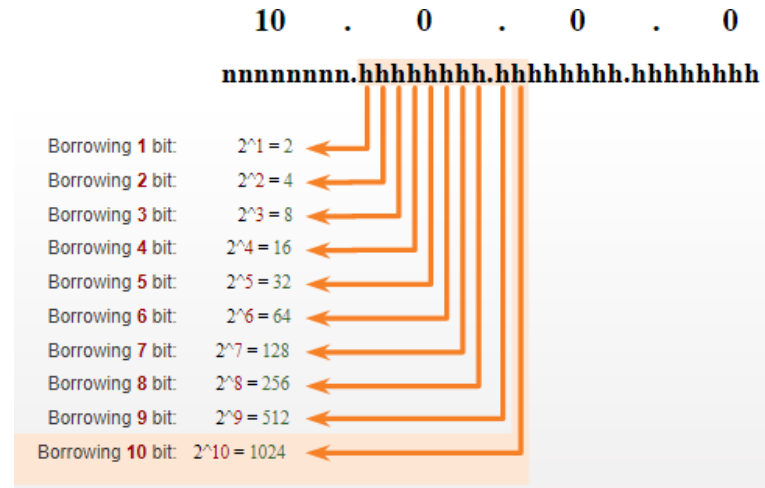
Subnet a Slash 16 and a Slash 8 Prefix

Create 1000 Subnets with a Slash 8 prefix

Consider a small ISP that requires 1000 subnets for its clients using network address 10.0.0.0/8 which means there are 8 bits in the network portion and 24 host bits available to borrow toward subnetting.

- The figure displays the number of subnets that can be created when borrowing bits from the second and third.
- Notice there are now up to 22 host bits that can be borrowed (i.e., last two bits cannot be borrowed).

To satisfy the requirement of 1000 subnets for the enterprise, 10 bits (i.e., $2^{10}=1024$ subnets) would need to be borrowed (for a total of 128 subnets)



Subnet a Slash 16 and a Slash 8 Prefix

Video – Subnet Across Multiple Octets

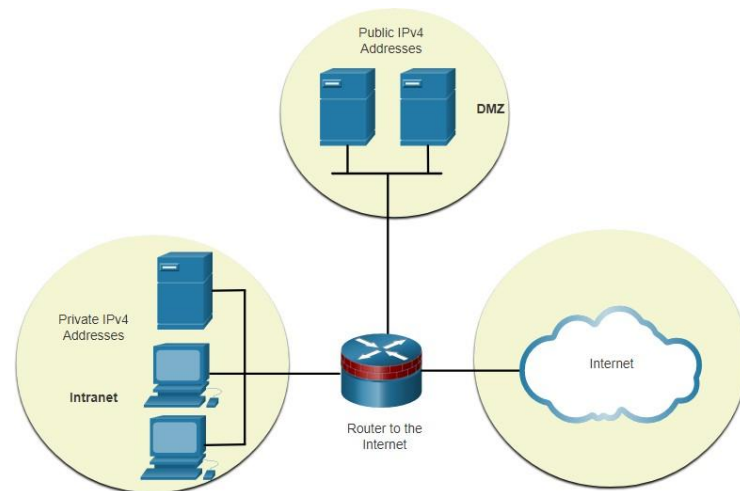


Subnet to Meet Requirements

Subnet Private versus Public IPv4 Address Space

Enterprise networks will have an:

- Intranet - A company's internal network typically using private IPv4 addresses.
- DMZ – A company's internet facing servers. Devices in the DMZ use public IPv4 addresses.
- A company could use the 10.0.0.0/8 and subnet on the /16 or /24 network boundary.
- The DMZ devices would have to be configured with public IP addresses.



Subnet to Meet Requirements

Minimize Unused Host IPv4 Addresses and Maximize Subnets

There are two considerations when planning subnets:

- The number of host addresses required for each network
- The number of individual subnets needed

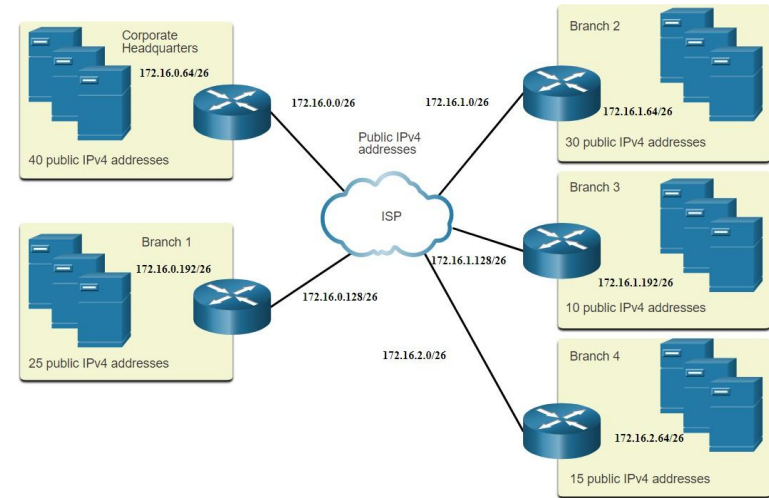
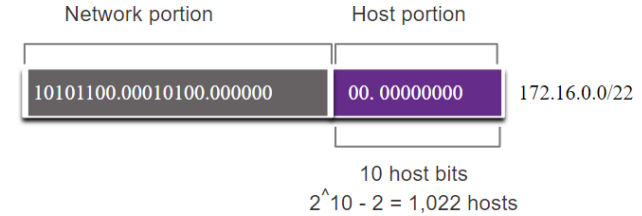


Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of subnets	# of hosts
/25	255.255.255.128	nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.nnnnnnnnn 11111111.11111111.11111111.10000000	2	126
/26	255.255.255.192	nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.nnnnnnnnn 11111111.11111111.11111111.11000000	4	62
/27	255.255.255.224	nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.nnnnnnnnn 11111111.11111111.11111111.11100000	8	30
/28	255.255.255.240	nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.nnnnnnnnn 11111111.11111111.11111111.11110000	16	14
/29	255.255.255.248	nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.nnnnnnnnn 11111111.11111111.11111111.11111000	32	6
/30	255.255.255.252	nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.nnnnnnnnn 11111111.11111111.11111111.11111100	64	2

Subnet to Meet Requirements

Example: Efficient IPv4 Subnetting

- In this example, corporate headquarters has been allocated a public network address of 172.16.0.0/22 (10 host bits) by its ISP providing 1,022 host addresses.
- There are five sites and therefore five internet connections which means the organization requires 10 subnets with the largest subnet requires 40 addresses.
- It allocated 10 subnets with a /26 (i.e., 255.255.255.192) subnet mask.

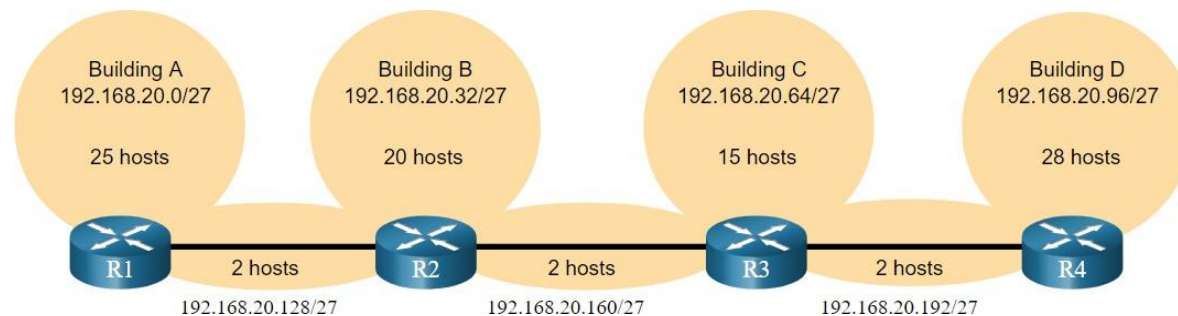


6.5 VLSM Calculation

IPv4 Address Conservation

Given the topology, 7 subnets are required (i.e, four LANs and three WAN links) and the largest number of host is in Building D with 28 hosts.

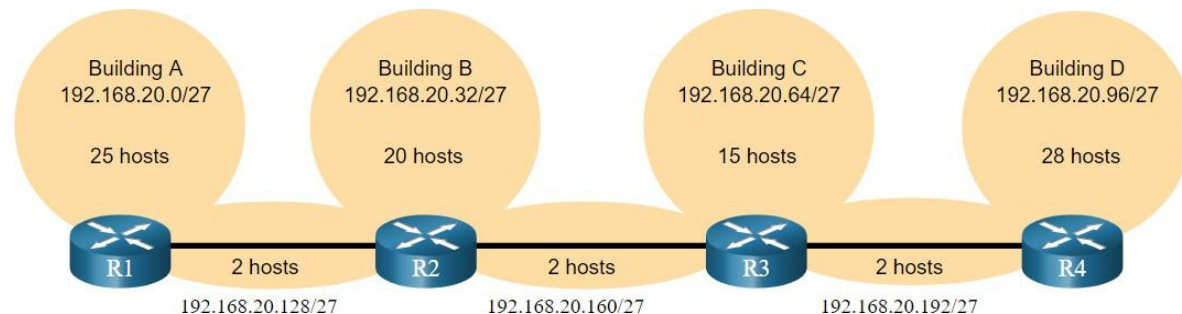
- A /27 mask would provide 8 subnets of 30 host IP addresses and therefore support this topology.



IPv4 Address Conservation (Cont.)

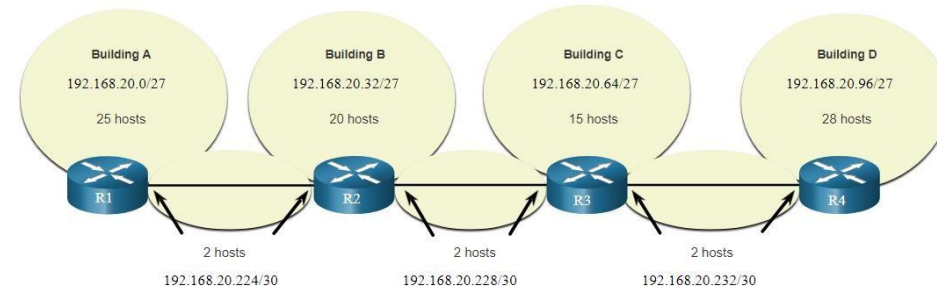
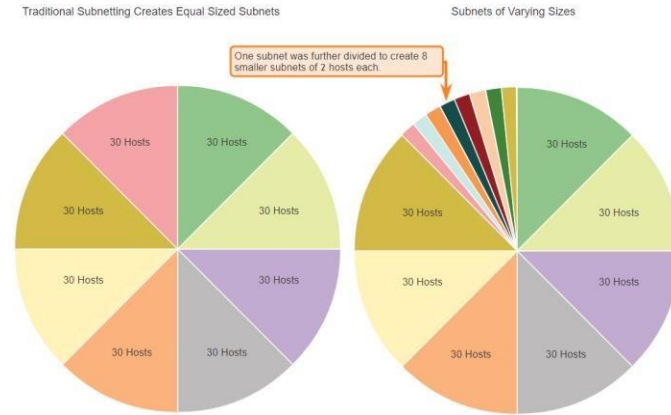
However, the point-to-point WAN links only require two addresses and therefore waste 28 addresses each for a total of 84 unused addresses.

Host portion
 $2^5 - 2 = 30$ host IP addresses per subnet
 $30 - 2 = 28$
Each WAN subnet wastes 28 addresses
 $28 \times 3 = 84$
84 addresses are unused



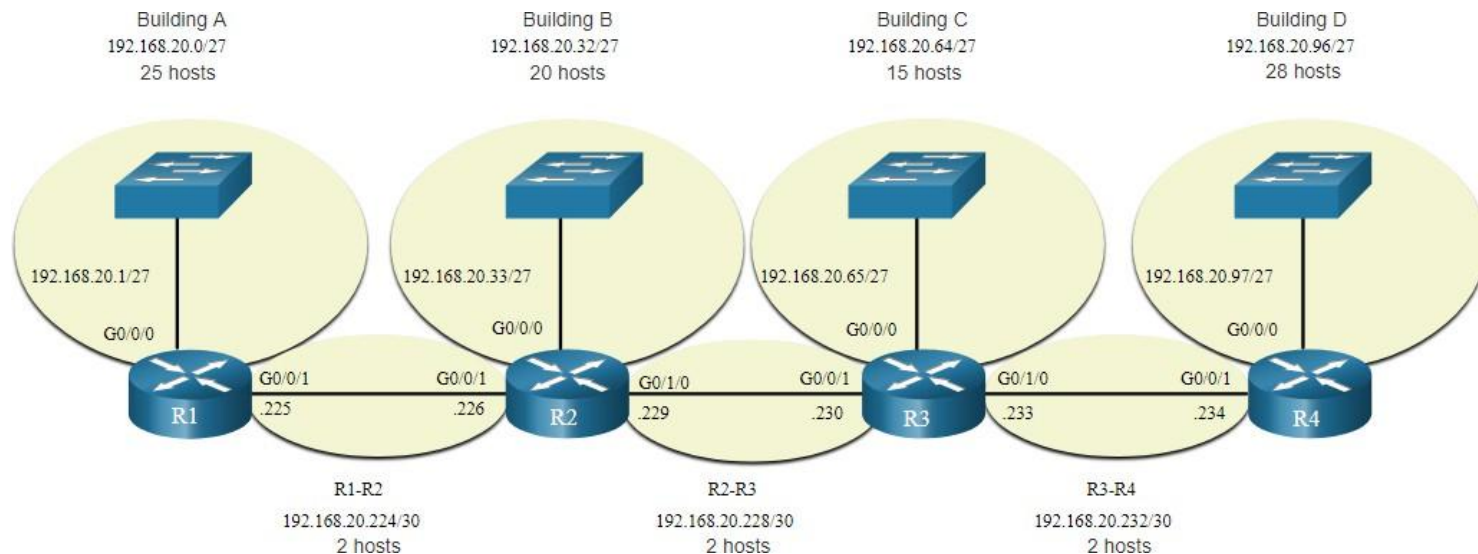
- Applying a traditional subnetting scheme to this scenario is not very efficient and is wasteful.
- VLSM was developed to avoid wasting addresses by enabling us to subnet a subnet.

- The left side displays the traditional subnetting scheme (i.e., the same subnet mask) while the right side illustrates how VLSM can be used to subnet a subnet and divided the last subnet into eight /30 subnets.
- When using VLSM, always begin by satisfying the host requirements of the largest subnet and continue subnetting until the host requirements of the smallest subnet are satisfied.
- The resulting topology with VLSM applied.



VLSM Topology Address Assignment

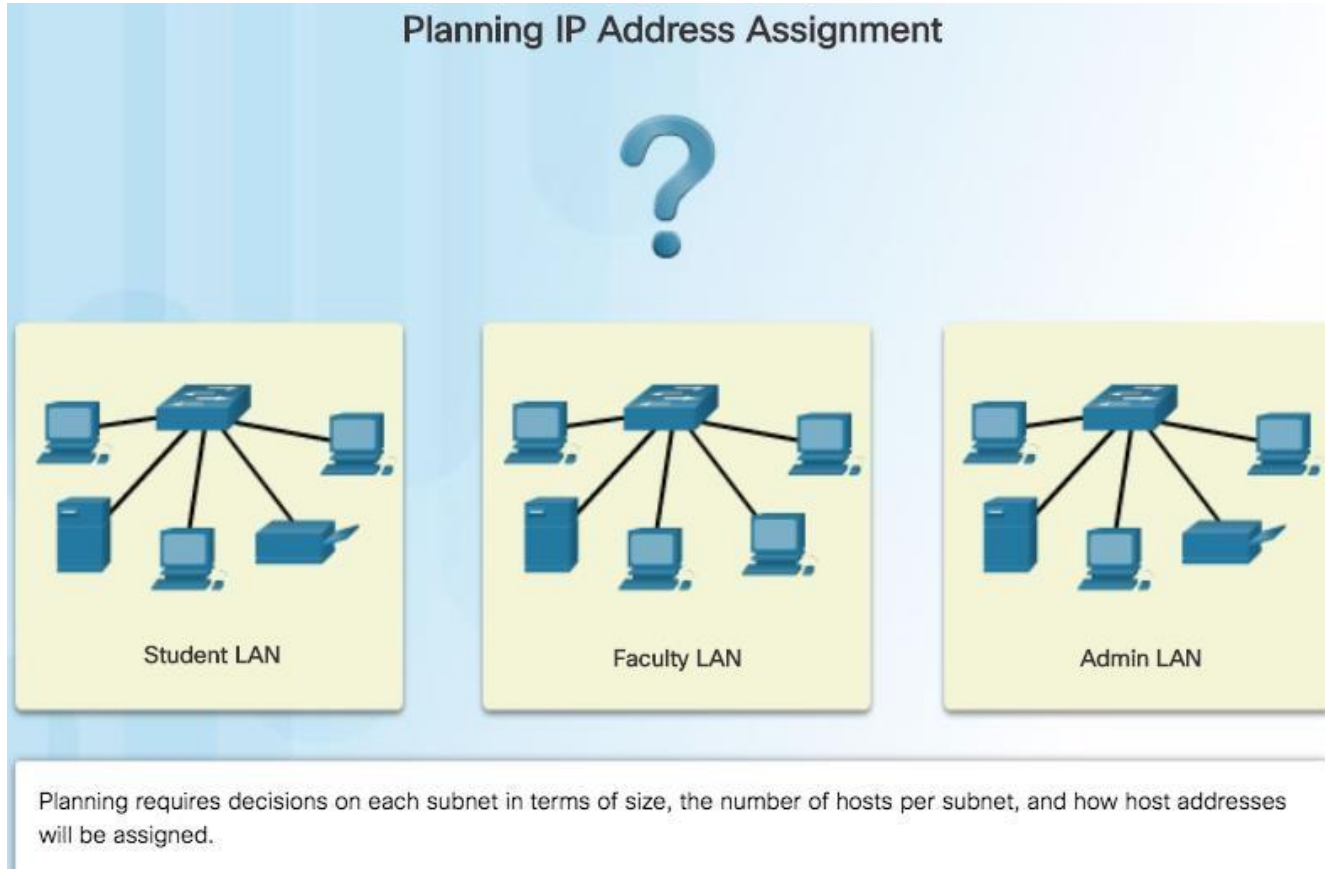
- Using VLSM subnets, the LAN and inter-router networks can be addressed without unnecessary waste as shown in the logical topology diagram.



Structured Design

Structured Design

Network Address Planning



Structured Design

IPv4 Network Address Planning

IP network planning is crucial to develop a scalable solution to an enterprise network.

- To develop an IPv4 network wide addressing scheme, you need to know how many subnets are needed, how many hosts a particular subnet requires, what devices are part of the subnet, which parts of your network use private addresses, and which use public, and many other determining factors.

Examine the needs of an organization's network usage and how the subnets will be structured.

- Perform a network requirement study by looking at the entire network to determining how each area will be segmented.
- Determine how many subnets are needed and how many hosts per subnet.
- Determine DHCP address pools and Layer 2 VLAN pools.

Structured Design

Device Address Assignment

Within a network, there are different types of devices that require addresses:

- **End user clients** – Most use DHCP to reduce errors and burden on network support staff. IPv6 clients can obtain address information using DHCPv6 or SLAAC.
- **Servers and peripherals** – These should have a predictable static IP address.
- **Servers that are accessible from the internet** – Servers must have a public IPv4 address, most often accessed using NAT.
- **Intermediary devices** – Devices are assigned addresses for network management, monitoring, and security.
- **Gateway** – Routers and firewall devices are gateway for the hosts in that network.

When developing an IP addressing scheme, it is generally recommended that you have a set pattern of how addresses are allocated to each type of device.

Network: 192.168.1.0/24		
Use	First	Last
Host Devices	.1	.229
Servers	.230	.239
Printers	.240	.249
Intermediary Devices	.250	.253
Gateway (router LAN interface)	.254	

